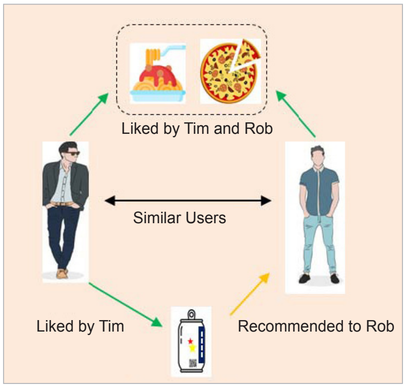

Figure 5: Collaborative filtering

These ratings might be considered as an approximation of the user's interest in the topic.
- The system matches this user's preferences against other user preferences and finds the people with the most 'similar' choice.
- The system recommends items that similar users have rated the most but have not yet been rated by this user (the absence of rating is often considered as unfamiliarity with an item).

The main challenge of collaborative filtering is combining and weighing the preferences of user neighbours. Users can sometimes rate the recommended items right away. As aresult, over time, the system obtains a more accurate depiction of consumer preferences.

The math behind finding the 'similarity' to recommend

So, how does a recommender system determine 'similarity'? The distance metric is used to do this. Similarity is a subjective concept that varies greatly depending on the domain and application. The points closest to each other are the most similar, while the points furthest apart are the least relevant. Two films, for example, may be comparable in terms of genre, length, or actors. When measuring distance between unrelated dimensions/ features, caution is advised. The relative values of each element must be normalised, else the distance calculation will be dominated by one feature.

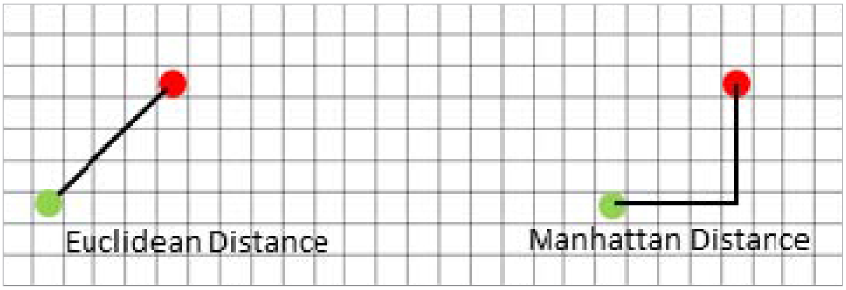

Figure 6: Euclidean distance and Manhattan distance

Minkowski distance: The Minkowski distance is the general form when the dimension of a data point is numeric. It's a metric for vector spaces with real values. Minkowski distance can only be calculated in a normed vector space, which is a space where distances can be represented as a vector with a length that cannot be negative. Manhattan(r=1) and Euclidean(r=2) distance measurements are generalisations of this generic distance metric.

$$\left(\sum_{i=1}^{n} |x_i - y_i|^p\right)^{1/p}$$

The formula shown above for Minkowski distance is in a generalised form, and we can manipulate it to get different distance metrics. Here, the 'p' value can be manipulated to give us different distances like:
- p = 1; when p is set to 1, it is Manhattan distance
- p = 2; when p is set to 2, it is Euclidean distance

Manhattan distance: This distance is also known as taxicab distance or city block distance because of how it is calculated. The sum of the absolute differences of two places' Cartesian coordinates is the distance between them – it's the distance between two places when measured at right angles along axes. The formula is:

$$\left(\sum_{i=1}^{n} |x_i - y_i|^p\right)$$

Euclidean distance: The square root of the sum of squares of the difference between the coordinates is given by the Pythagorean theorem:

$$d(x, y) = \sqrt{\sum_{i}^{n} (x_i - y_i)^2}$$

Cosine similarity: This is a metric that is useful in determining how similar the data objects are, irrespective of their size. Cosine similarity calculates the cosine of the angle between two vectors to determine how similar they are. In cosine similarity, data objects in a data set are treated as a vector. The formula to find the cosine similarity between two vectors is:

```
Cos(x, y) = x . y / ||x|| *
||y||
```

where,
- x, y = product (dot) of the vectors 'x' and 'y'
- ||x|| and ||y|| = length of the two vectors 'x' and 'y'
- ||x|| * ||y|| = cross product of the two vectors 'x' and 'y'

The cosine similarity between two vectors is measured in 'θ'. If θ = 0°, the 'x' and 'y' vectors overlap, thus proving they are similar. If θ = 90°, the 'x' and 'y' vectors are dissimilar.

Word embedding in the context of building a recommender system

Word embeddings are texts that have been translated into numbers, and there may be multiple numerical